Analytics & Big Data

Session 9: Big data II

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🎯 **Learning objectives**

Tentatively

- **Explain** the characteristics and challenges of unstructured data sources (e.g., text, social media, spatial data).
- **Acquire and preprocess** large-scale or unstructured data using APIs and appropriate tools in Python.
- **Apply** large language models or other scalable methods to extract, classify, or summarize unstructured data.
- **Evaluate** the reliability, bias, and limitations of API- and LLM-based analytics workflows.

TBD: spatial/social-media analytics?

TODO

Reinforcement Learning

The solution to many of the problems in our lives cannot be automated. This is not because current computers are too slow, but simply because it is too difficult for humans to determine what the program should do.

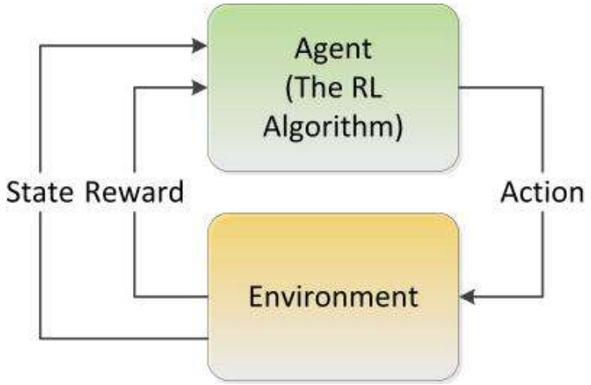
Supervised learning is a general method for training an approximator. However, supervised learning requires sample input-output pairs from the domain to be learned.

For example, we might not know the best way to program a computer to recognize an infrared picture of a tank, but we do have a large collection of infrared pictures, and we do know whether each picture contains a tank or not. Supervised learning could look at all the examples with answers, and learn how to recognize tanks in general.

Unfortunately, there are many situations where we don’t know the correct answers that supervised learning requires. For example, in a self-driving car, the question would be the set of all sensor readings at a given time, and the answer would be how the controls should react during the next millisecond.

For these cases there exist a different approach known as reinforcement learning.

Reinforcement Learning



> The agent learns how to achieve a given goal by trial-and-error interactions with its environment by maximizing a reward.

AlphaGo

Go is one of the hardest games in the world for AI because of the huge number of different game scenarios and moves. The number of potential legal board positions is greater than the number of atoms in the universe.

The core of AlphaGo is a deep neural network. It was initially trained to learn playing by using a database of around 30 million recorded historical moves. After the training, the system was cloned and it was trained further playing large numbers of games against other instances of itself, using reinforcement learning to improve its play. During this training AlphaGo learned new strategies which were never played by humans.

A newer version named AlphaGo Zero skips the step of being trained and learns to play simply by playing games against itself, starting from completely random play.

Libratus

An artificial intelligence called Libratus has beaten four of the world's best poker players in a grueling 20-day tournament in January 2017.

Poker is more difficult because it's a game with imperfect information. With chess and Go, each player can see the entire board, but with poker, players don't get to see each other's hands. Furthermore, the AI is required to bluff and correctly interpret misleading information in order to win.

"We didn't tell Libratus how to play poker. We gave it the rules of poker and said 'learn on your own'." The AI started playing randomly but over the course of playing trillions of hands was able to refine its approach and arrive at a winning strategy.

Types of Artificial Intelligence

Discriminative AI is designed to differentiate and classify input, but not to create new content. Examples include image or speech recognition, credit scoring or stock price prediction.

Generative AI is able to generate new content based on existing information and user specifications. This includes texts, images, videos, program code, etc. The generated content can often hardly be distinguished from human-generated content. As things stand at present, however, they are pure recombinations of learned knowledge.

Well-known examples of generative AI are language models for generating text, such as GPT-3 or GPT-4, and the chatbot ChatGPT based on them, or image generators such as Stable Diffusion and DALL-E.

ChatGPT

ChatGPT is a generative AI that produces human-like text and communicates with humans.

The “GPT” in ChatGPT comes from the language model of the same name, which was extended for ChatGPT with various components for communication and quality assurance.

GPT is based on a huge neural network that essentially represents the language model. While the first GPT-3 has 175 billion parameters, the newer GPT-4 already has 1 trillion parameters. Compared to GPT-3, GPT-4 is therefore more intelligent, can deal with more extensive questions and conversations and makes fewer factual errors.

ChatGPT - Approach

ChatGPT generates its response word by word via a sequence of probabilities, with each new word depending on the previous ones.

The best thing about AI is its ability to

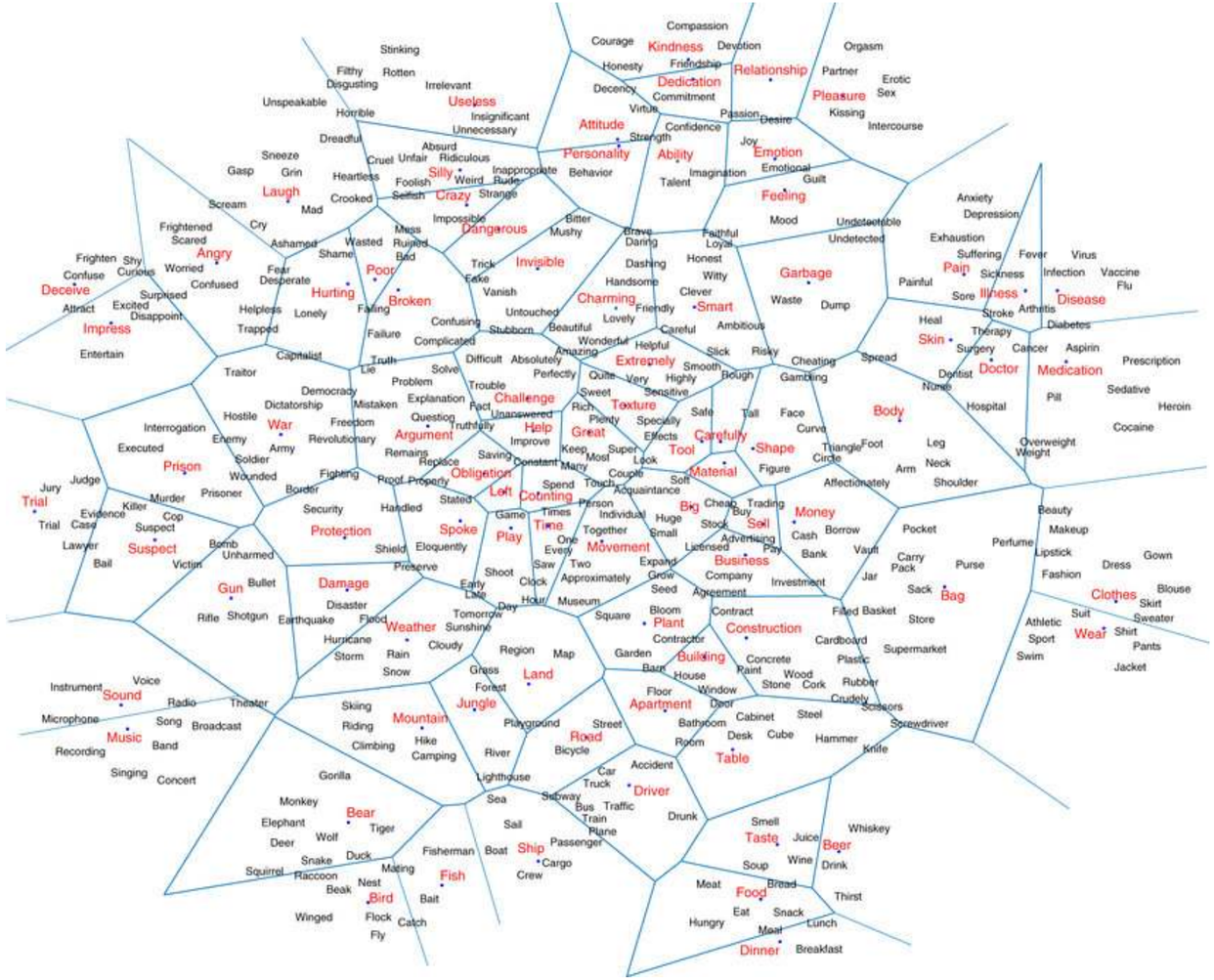
learn	4.5%
predict	3.5%
make	3.2%
understand	3.1%
do	2.9%

The best thing about AI is its ability to,
The best thing about AI is its ability to learn,
The best thing about AI is its ability to learn from,
The best thing about AI is its ability to learn from experience,
The best thing about AI is its ability to learn from experience.,
The best thing about AI is its ability to learn from experience. It,
The best thing about AI is its ability to learn from experience. It's,
The best thing about AI is its ability to learn from experience. It's not

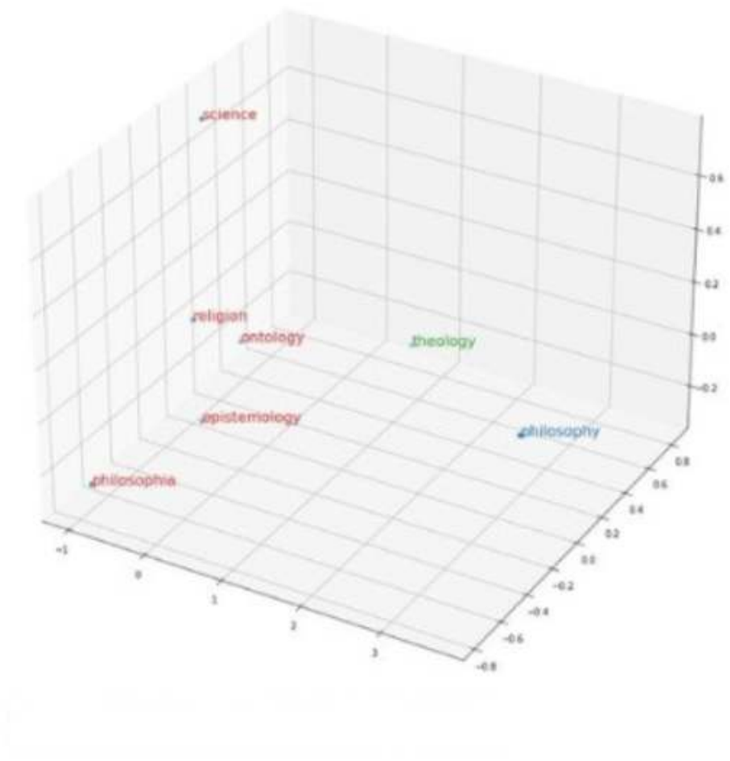
The most probable word is not always selected; instead, randomization takes place. This means that different variants can be created for the same task.

The best thing about AI is its ability to learn. I've always liked the
The best thing about AI is its ability to really come into your world and just
The best thing about AI is its ability to examine human behavior and the way it
The best thing about AI is its ability to do a great job of teaching us
The best thing about AI is its ability to create real tasks, but you can

ChatGPT - Semantic Spaces (I)


Compassion
Courage
Kindness
Devotion
Orgasm
Honesty
Friendship
Relationship
Partner
Erotic
Decency
Dedication
Commitment
Pleasure
Sex
Virtue
Passion
Desire
Kissing
Intercourse
Attitude
Confidence
Strength
Joy
Personality
Ability
Emotion
Talent
Imagination
Emotional
Guilt
Feeling
Mood
Undetectable
Undetected
Anxiety
Depression
Exhaustion
Suffering
Fever
Virus
Pain
Sickness
Infection
Vaccine
Flu
Illness
Disease
Stroke
Arthritis
Diabetes
Heal
Therapy
Skin
Surgery
Cancer
Aspirin
Doctor
Medication
Prescription
Dentist
Nurse
Hospital
Pill
Sedative
Heroin
Cocaine
Overweight
Weight
Body
Leg
Neck
Shoulder
Arm
Foot
Triangle
Circle
Affectionately
Shape
Figure
Tool
Carefully
Material
Face
Curve
Tall
Safe
Texture
Great
Help
Improve
Keep
Super
Most
Look
Couple
Soft
Spend
Touch
Acquaintance
Person
Big
Cheap
Trading
Buy
Money
Cash
Borrow
Pocket
Soil
Stock
Huge
Small
Advertising
Pay
Bank
Vault
Carry
Pack
Purse
Perfume
Beauty
Makeup
Lipstick
Dress
Gown
Fashion
Clothes
Blouse
Skirt
Sweater
Suit
Wear
Shirt
Pants
Jacket
Athletic
Sport
Swim
Bag
Sack
Jar
Investment
Business
Licensed
Company
Agreement
Contract
Filled Basket
Store
Supermarket
Cardboard
Construction
Contractor
Plant
Square
Bloom
Building
Garden
Barn
House
Concrete
Paint
Wood
Stone
Cork
Rubber
Plastic
Crudely
Scissors
Screwdriver
Knife
Hammer
Cube
Steel
Cabinet
Desk
Room
Table
Apartment
Floor
Window
Door
Bathroom
Street
Road
Bicycle
Accident
Driver
Car
Truck
Bus
Traffic
Train
Plane
Subway
Passenger
Cargo
Crew
Ship
Boat
Fisherman
Sail
Sea
Lighthouse
River
Map
Region
Grass
Forest
Land
Playground
Jungle
Mountain
Hike
Camping
Climbing
Riding
Skiing
Theater
Broadcast
Radio
Voice
Instrument
Sound
Microphone
Song
Music
Band
Recording
Singing
Concert
Gorilla
Monkey
Bear
Tiger
Elephant
Wolf
Deer
Snake
Duck
Squirrel
Raccoon
Beak
Nest
Mating
Fish
Bait
Bird
Flock
Catch
Winged
Fly
Drunk
Smell
Taste
Juice
Whiskey
Beer
Wine
Drink
Soup
Meat
Food
Bread
Eat
Snack
Lunch
Hungry
Meal
Breakfast
Dinner
Thirst
Stinking
Filthy
Rotten
Disgusting
Irrelevant
Unspeakable
Useless
Insignificant
Unnecessary
Horrible
Dreadful
Absurd
Sneeze
Cruel
Unfair
Ridiculous
Gasp
Grin
Heartless
Silly
Inappropriate
Behavior
Foolish
Weird
Rude
Laugh
Crooked
Selfish
Crazy
Strange
Scream
Mad
Impossible
Frightened
Cry
Mess
Dangerous
Bitter
Mushy
Scared
Ashamed
Wasted
Ruined
Shame
Bad
Angry
Frighten
Shy
Trick
Invisible
Brave
Daring
Faithful
Loyal
Dashing
Honest
Confuse
Curious
Worried
Fake
Poor
Fear
Desperate
Confused
Deceive
Surprised
Vanish
Hurting
Handsome
Garbage
Clever
Witty
Attract
Excited
Disappoint
Helpless
Lonely
Falling
Broken
Charming
Friendly
Smart
Waste
Dump
Impress
Trapped
Untouched
Confusing
Lovely
Careful
Ambitious
Entertain
Failure
Stubborn
Beautiful
Complicated
Wonderful
Helpful
Slick
Risky
Cheating
Spread
Capitalist
Truth
Amazing
Extremely
Smooth
Gambling
Traitor
Lie
Difficult
Absolutely
Rough
Solve
Perfectly
Quite
Very
Highly
Sensitive
Democracy
Problem
Trouble
Sweet
Dictatorship
Mistaken
Explanation
Challenge
Hostile
Freedom
Question
Fact
Unanswered
Rich
War
Truthfully
Plenty
Specially
Interrogation
Revolutionary
Argument
Effects
Enemy
Remains
Improve
Safe
Tall
Face
Executed
Army
Replace
Saving
Constant
Many
Soldier
Prison
Wounded
Obligation
Proof
Property
Fighting
Jury
Judge
Trial
Murder
Prisoner
Border
Stated
Loft
Counting
Murder
Evidence
Killer
Cop
Case
Suspect
Security
Handled
Game
Times
Person
Individual
Protection
Spoke
Play
Time
Huge
Lawyer
Bail
Bomb
Unharmed
Victim
Shield
Eloquently
Every
One
Together
Small
Movement
Preserve
Shoot
Saw
Two
Expand
Licensed
Business
Gun
Bullet
Damage
Early
Clock
Approximately
Grow
Company
Investment
Seed
Late
Hour
Museum
Agreement
Rifle
Shotgun
Disaster
Earthquake
Flood
Tomorrow
Square
Bloom
Contract
Weather
Sunshine
Plant
Hurricane
Cloudy
Contractor
Storm
Rain
Region
Map
Garden
Snow
Grass
Land
Barn
House
Floor
Window
Forest
Jungle
Apartment
Door
Mountain
Playground
Street
Bathroom
Cabinet
Road
Bicycle
Room
Desk
Cube
Table
Hike
Camping
River
Accident
Lighthouse
Car
Truck
Driver
Sea
Subway
Bus
Traffic
Train
Plane
Sail
Passenger
Ship
Boat
Fisherman
Cargo
Crew
Fish
Bait
Bird
Flock
Catch
Fly
Winged
Nest
Beak
Raccoon
Mating
Duck
Snake
Squirrel
Deer
Wolf
Elephant
Tiger
Bear
Monkey
Gorilla
Theater
Broadcast
Radio
Voice
Instrument
Sound
Microphone
Song
Music
Band
Recording
Singing
Concert
Skiing
Riding
Climbing
Hike
Camping
Lighthouse
Sea
Sail
Fisherman
Boat
Ship
Passenger

ChatGPT - Semantic Spaces (II)

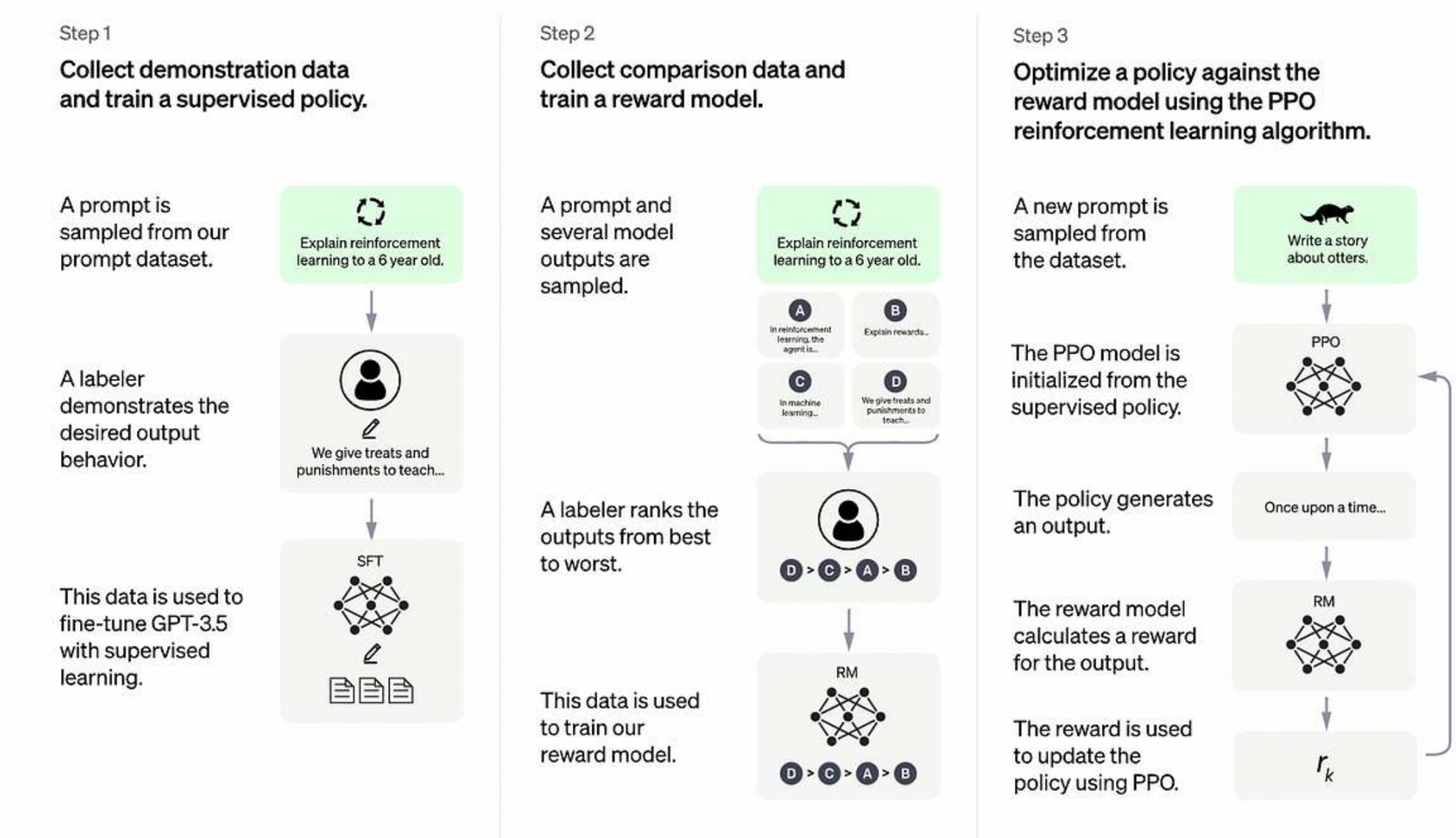



```
print_neighbors('ontology', model5)
```

Approximate Neighbors

```
('Eros', 0.8966988027095795)
('Astronomy', 0.8926706984639168)
('Philosophy', 0.886092908680439)
('mathematics', 0.8845213651657104)
('Psychology', 0.8803791999816895)
('Night', 0.8794359490275383)
('subject-object', 0.8780221417546272)
('king', 0.8750540241599083)
('epistemology', 0.8732230812311172)
('Dionysus', 0.8667336702346802)
```

ChatGPT - Evaluation Component



Survey: Session 9

https://forms.gle/1vsCsqc3SzWfSX1f6

Note: Responses may be analyzed and published in anonymized form.

Please complete the survey before you leave today — thank you 🙏

References